

Bayesian Hierarchical Prediction of Used Car Prices

Group Members

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Abstract

Used car prices are influenced by many vehicle characteristics and can also vary across brands. We build a Bayesian hierarchical regression model for log price, using vehicle level predictors and brand level random intercepts to partially pool information across brands. The model was fit in JAGS and compared with an OLS baseline. OLS produced slightly lower test RMSE, while the Bayesian model gave posterior predictive intervals with about 94.5% coverage.

1. Introduction

Used car prices are influenced by many observable vehicle characteristics, including mileage, model year, fuel type, transmission type, accident history, title status, and brand. Accurately predicting used car prices is useful for both buyers and sellers because it can help consumers evaluate whether a listed price is reasonable and help sellers understand how different vehicle features affect market value. However, used car prices can vary substantially across brands, even after accounting for individual vehicle characteristics. For example, luxury brands may have systematically higher baseline prices than non-luxury brands.

The goal of this project is to build and evaluate a Bayesian hierarchical prediction model for used car prices using a real world dataset of used vehicle listings. We model the log of vehicle price as the response variable and use vehicle level predictors such as age, mileage, fuel type, transmission, accident history, and clean title status. To account for systematic differences across vehicle brands, we include brand level random intercepts in the model. This hierarchical structure allows different brands to have different baseline price levels while partially pooling information across brands.

2. Data and Preprocessing

The data used in this project come from the Used Car Price Prediction Dataset on Kaggle, created by Taeef Najib (Najib). The dataset contains information on used vehicle listings, including vehicle price, brand, model year, mileage, fuel type, transmission type, accident history, clean title status, and other vehicle characteristics. The original dataset contains 4,009 observations and 12 variables.

The response variable in this analysis is the listed vehicle price. Since vehicle prices are positive and strongly right skewed, we modeled the logarithm of price rather than the raw price. Mileage was also transformed using $\log(\text{mileage} + 1)$ to reduce skewness and the influence of very high mileage observations. Vehicle age was constructed from model year using 2024, the maximum model year in the cleaned dataset, as the reference year:

$$age_i = 2024 - model\ year_i.$$

Several preprocessing steps were performed before fitting the models. First, the price and mileage variables were converted from character strings into numeric variables by removing currency symbols, commas, and mileage units. Observations with impossible or invalid values were removed. To focus on mainstream used car listings and reduce the influence of extreme luxury or exotic vehicles, we excluded listings with prices above \$200,000 and mileage above 300,000 miles. After filtering, the cleaned dataset contained 3,928 observations.

Categorical predictors were also cleaned and simplified. Missing values in categorical variables were treated as an “Unknown” category when appropriate. The transmission variable originally contained many detailed labels, so it was simplified into broader categories: Automatic, Manual, CVT, Other, and Unknown. Rare vehicle brands were combined into an “Other” category to avoid sparse group level estimates in the hierarchical model. After this step, the final brand grouping variable contained 35 levels. The other categorical predictors included fuel type with 6 levels, transmission with 5 levels, accident history with 3 levels, and clean title status with 2 levels.

For model fitting, the cleaned data were split into training and testing sets using an 80/20 split stratified by brand group. This produced 3,143 training observations and 785 testing observations. The training set was used to fit both the Bayesian hierarchical model and the OLS regression model, while the testing set was used to evaluate out-of-sample predictive performance. Continuous predictors were standardized using the training set mean and standard deviation before model fitting. The final fixed effect design matrix contained 14 predictor columns after encoding categorical variables as dummy variables.

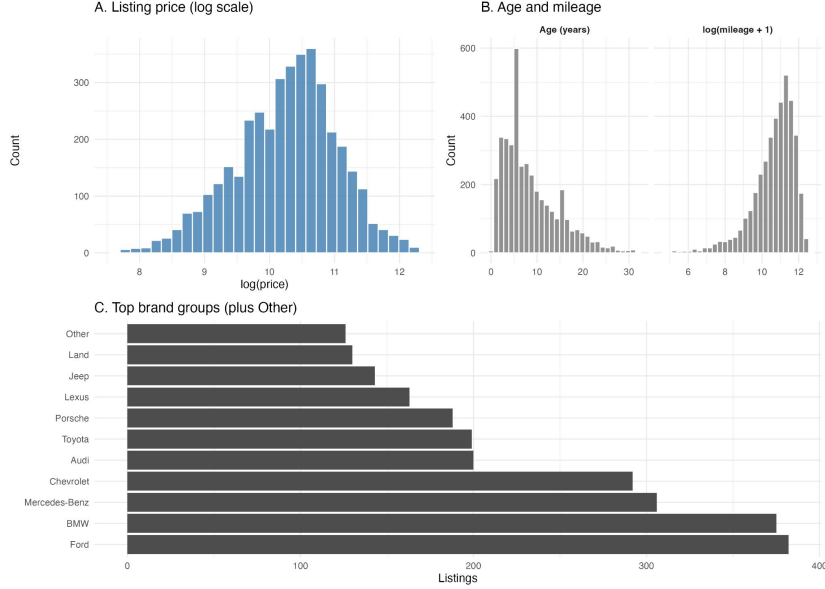
Table 1 Summary of variables and preprocessing steps

Variable	Role	Type	Preprocessing
Price	Response	Cont.	Transformed to (log(Price))
Model Year	Predictor	Numeric	Converted to vehicle age
Mileage	Predictor	Cont.	Transformed to (log(mileage+1))
Brand	Grouping variable	Cat.	Rare brands combined; 35 final levels
Fuel Type	Predictor	Cat.	Missing/rare values grouped; 6 levels
Transmission	Predictor	Cat.	Collapsed into 5 broad categories
Accident History	Predictor	Cat.	3 levels including Unknown
Clean Title	Predictor	Cat.	2 levels

We also conduct a simple exploratory data overview after the data cleaning process. Figure 1 below summarizes the cleaned sample of 3,928 listings. Panel A shows that log listing price is roughly unimodal, which supports modeling log price rather than raw dollars. Panel B shows that vehicle age is concentrated in recent model years and that log (mileage + 1) is right skewed on the original mileage

scale, consistent with the transforms in Table 1. Panel C shows uneven listing counts across the largest brand groups and the ‘Other’ category formed by pooling rare brands, motivating brand level random intercepts with partial pooling and the brand stratified train test split described above.

Figure 1. Exploratory data overview of the cleaned used car listings



3. Bayesian Hierarchical Model

We use a Bayesian hierarchical regression model to predict used car prices. Since the original price variable is positive and right-skewed, the response variable is defined as the centered log price:

$$y_i = \log(\text{price}_i) - \overline{\log(\text{price})}_{\text{train}}$$

where $i = 1, \dots, n$ indexes vehicles in the training set. Centering the log price helps improve computation and makes the intercept easier to model.

For each vehicle i , we assume

$$y_i | \mu_i, \sigma^2 \sim N(\mu_i, \sigma^2),$$

where μ_i is the expected centered log price and σ^2 is the residual variance. The linear predictor is

$$\mu_i = \alpha_{b[i]} + x_i^T \beta,$$

where $b[i]$ denotes the brand group of vehicles i , $\alpha_{b[i]}$ is the brand-specific random intercept, x_i is the vector of fixed effect predictors, and β is the corresponding vector of regression coefficients. The fixed effect predictors include standardized vehicle age, standardized log mileage, fuel type, transmission type, accident history, and clean title status. After dummy encoding the categorical variables, the design matrix contained $p = 14$ fixed-effect columns.

To account for systematic price differences across vehicle brands, we introduce a hierarchical structure through brand level random intercepts. If there are $J = 35$ brand groups, then for each brand group $j = 1, \dots, J$,

$$\alpha_j | \mu_\alpha, \sigma_\alpha^2 \sim N(\mu_\alpha, \sigma_\alpha^2).$$

Here, μ_α represents the overall mean brand level intercept, and σ_α^2 measures the amount of variation in baseline log prices across brands. This structure allows different brands to have different baseline price levels while partially pooling information across brands. Brands with fewer observations are pulled toward the overall mean, which can help stabilize estimates for smaller groups.

We assign weakly informative priors to the model parameters. For the fixed effect coefficients,

$$\beta_k \sim N(0, 2^2), k = 1, \dots, p.$$

For the overall brand level mean,

$$\mu_\alpha \sim N(0, 5^2).$$

For the residual standard deviation and the brand level standard deviation, we use weakly informative half normal priors:

$$\sigma \sim \text{Half} - \text{Normal}(0, 2^2), \sigma_\alpha \sim \text{Half} - \text{Normal}(0, 2^2).$$

These priors are weakly informative because the response variable is centered log price and the continuous predictors are standardized. Therefore, values of the regression coefficients or standard deviations much larger than 2 would represent very large changes on the log price scale. The priors provide regularization while still allowing the data to determine the main posterior patterns.

A prior sensitivity check replaced $\beta_k \sim N(0, 2^2)$ with $\beta_k \sim N(0, 1^2)$, while keeping the priors on μ_α , σ , and σ_α unchanged. Posterior means changed only slightly and test RMSE/MAE changed by less than one percent, so the baseline $\beta_k \sim N(0, 2^2)$ prior was retained.

Prior sensitivity: posterior mean comparison

Parameter	Baseline mean	Alternative mean	Difference
σ	0.43683	0.43684	5.07e-06
σ_α	0.34505	0.34605	0.000999
μ_α	0.29593	0.29444	-0.00149
β_1	-0.36313	-0.36280	0.000324
β_2	-0.28217	-0.28225	-0.0000758
β_3	-0.54236	-0.53822	0.00414

Prior sensitivity: test-set prediction metrics

Prior	Test RMSE	Test MAE	Diff. RMSE	Diff. MAE
$\beta_k \sim N(0, 2^2)$ baseline	24,075	13,431	—	—
$\beta_k \sim N(0, 1^2)$ alternative	24,024	13,428	-51	-3

The model was estimated using Markov chain Monte Carlo methods implemented in JAGS. In JAGS, normal distributions are parameterized using precision rather than variance, so we used

$$\tau = \frac{1}{\sigma^2}, \tau_\alpha = \frac{1}{\sigma_\alpha^2}.$$

Thus, the hierarchical model combines vehicle level predictors with brand level random effects to produce posterior predictive distributions for used car prices.

4. MCMC Computation and Diagnostics

The Bayesian hierarchical model was estimated using Markov chain Monte Carlo methods implemented in JAGS through the rjags package in R. The model was run with three parallel chains. We used 1,000 adaptation iterations, followed by 5,000 burn-in iterations. After burn-in, we drew 10,000 iterations from each chain with a thinning interval of 5, resulting in 2,000 posterior samples per chain and 6,000 posterior samples in total.

The parameters monitored in the MCMC sampling included the fixed effect regression coefficients β , the brand specific random intercepts α_j , the overall brand level mean μ_α , the residual standard deviation σ , and the brand level standard deviation σ_α . These parameters were used to summarize the posterior distribution and to generate posterior predictive distributions for the test data.

We assessed convergence using trace plots, Gelman-Rubin diagnostics, and effective sample sizes. The trace plots for the main hyperparameters and selected regression coefficients showed stable mixing across the three chains, with no obvious trends or separation between chains. The Gelman-Rubin point estimates were close to 1 for all monitored parameters, mostly between 1.00 and 1.02, suggesting no major convergence problems. Effective sample sizes were also generally adequate for posterior inference and prediction.

Monte Carlo error was assessed with coda time series standard errors on saved baseline draws for σ , σ_α , μ_α , and $\beta[1] \sim \beta[3]$. MCSE to posterior SD ratios ranged from about 0.013 to 0.039, with ESS roughly 640-6,300. So, Monte Carlo error was minor relative to posterior uncertainty.

MCMC MCSE summary

Parameter	Posterior mean	Posterior SD	ESS	MCSE	MCSE / SD
σ	0.43683	0.005582	6,123.58	0.0000714	0.0128
σ_α	0.34505	0.046686	6,000.00	0.0006028	0.0129
μ_α	0.29593	0.082373	767.47	0.0030122	0.0366
β_1	-0.36313	0.010873	5,766.83	0.0001433	0.0132
β_2	-0.28217	0.010518	5,302.16	0.0001447	0.0138
β_3	-0.54236	0.063379	631.86	0.0025364	0.0400

Because the MCMC chains appeared to mix well and the posterior summaries did not drift in ways that would suggest non-convergence, we treated the JAGS output as a reliable approximation to the posterior. On that basis, we compared how well two closely related models fit the training data using standard predictive summaries DIC, WAIC, and LOO that appear in Section 5. Both models use the same likelihood for (centered) log price and the same training observations; the only deliberate change in the sensitivity analysis was the prior on the regression coefficients beta: a more diffuse N (0, 4) versus a tighter N (0, 1) specification (matching the variance interpretation implied by the JAGS

precision parameterization in the model files). DIC came from `dic.samples` output saved during the JAGS run; WAIC and LOO were computed in R from the saved draws by forming, for each posterior draw and each training case, the corresponding Normal log likelihood contribution, then stacking these into an S by n matrix for the `loo` package. Across these criteria, the two priors led to nearly identical scores, which suggests that substantive conclusions are not sensitive to that particular prior choice. We therefore report posterior inference and held out test predictions from the baseline model throughout the remainder of the project.

Figure 2 Trace plots for the overall brand level mean

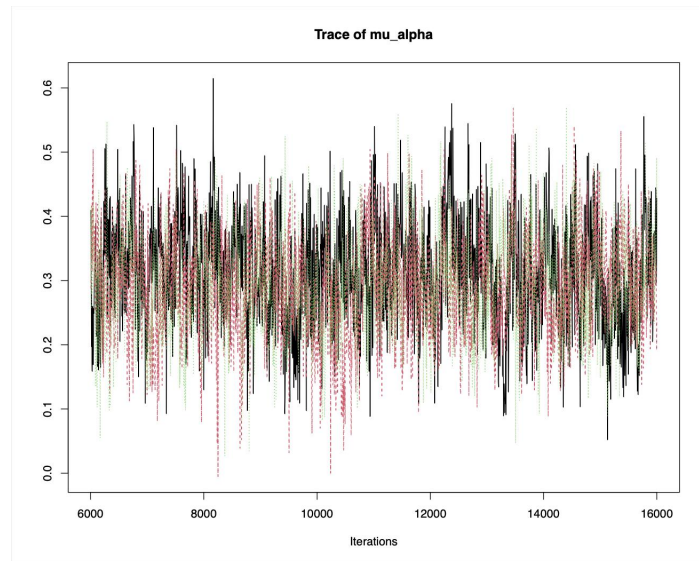
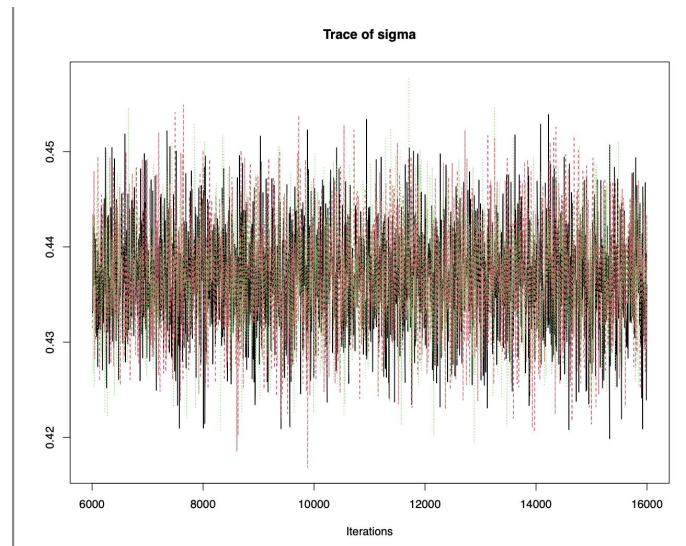


Figure 3 Trace plot for the residual standard deviation σ



Figures 2 and 3 show trace plots for two main hyperparameters in the Bayesian hierarchical model: the overall brand level mean μ_α and the residual standard deviation σ . The three MCMC chains appear to mix well and fluctuate around stable ranges without clear upward or downward trends. This suggests that the chains reached a stable posterior distribution.

5. Prediction Results and Model Comparison

After fitting the Bayesian hierarchical model, we evaluated its predictive performance on the held out test set of 785 observations. For each test observation, posterior predictive draws were generated using the sampled model parameters. These predictions were then transformed back to the original dollar price scale. We summarized point prediction accuracy using root mean squared error (RMSE) and mean absolute error (MAE).

To evaluate whether the Bayesian hierarchical structure improved prediction, we compared it with a standard ordinary least squares (OLS) regression model. The OLS model used the same training and testing split and included the same vehicle level predictors, along with brand indicators. Therefore, the comparison focuses on whether the Bayesian hierarchical partial pooling structure improves out-of-sample prediction relative to an ordinary regression approach.

The prediction results are shown in Table 2 below.

Table 2 . Test set prediction performance for OLS and Bayesian hierarchical models

Model	Test RMSE	Test MAE
OLS regression	23,057.40	13,013.17
Bayesian hierarchical model	24,052.79	13,429.06

On the original price scale, the OLS regression model achieved a slightly lower test RMSE and MAE than the Bayesian hierarchical model. Specifically, the OLS model had a test RMSE of 23,057.40 and a test MAE of 13,013.17, while the Bayesian hierarchical model had a test RMSE of 24,052.79 and a test MAE of 13,429.06. This indicates that, in this train test split, the OLS model slightly outperformed the Bayesian hierarchical model in terms of point prediction accuracy.

However, the Bayesian hierarchical model also provides predictive uncertainty through posterior predictive intervals. The 95% posterior predictive intervals had a test set coverage of approximately 94.5%, which is close to the nominal 95% level. This suggests that although the Bayesian hierarchical model did not produce lower point prediction errors than OLS, it produced reasonably calibrated uncertainty estimates.

The Figure 4 and 5 also show that both models perform reasonably well for low and mid price vehicles, where most observations are concentrated. However, both models show larger prediction errors for high priced vehicles. This suggests that the available predictors capture much of the variation in ordinary used car prices but may not fully explain the pricing of expensive or unusual vehicles. Additional information such as trim level, detailed vehicle condition, location, and optional features could potentially improve prediction for these high price listings.

Figure 4 Actual versus predicted prices for the Bayesian hierarchical model

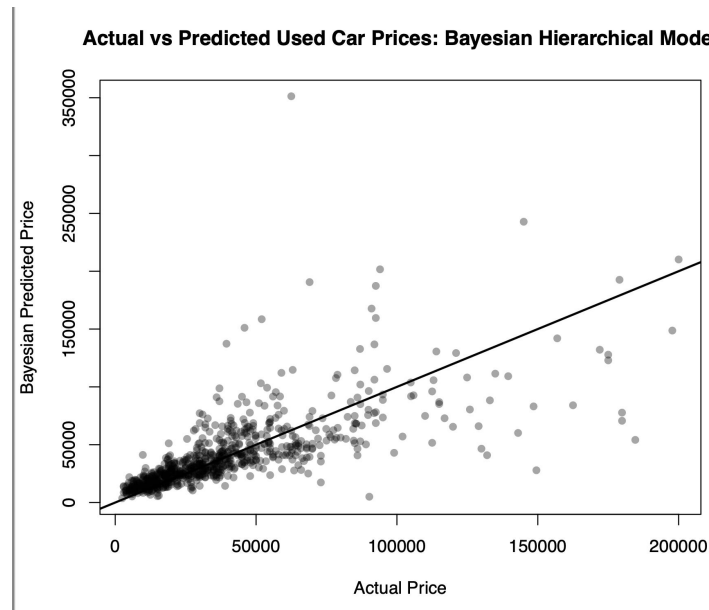
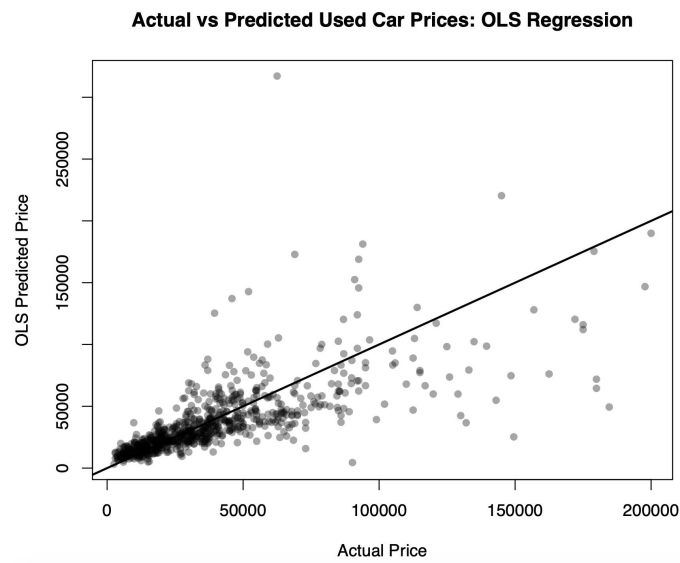


Figure 5 Actual versus predicted prices for the OLS regression model



Training set DIC was about 3760 for both beta prior specifications. WAIC was 3759.9 for the baseline prior and 3759.5 for the tighter prior, while LOOIC was 3760.1 for the baseline prior and 3759.7 for the tighter prior. These differences are negligible, so the in sample criteria do not strongly prefer one beta prior. The test set ranking also remains unchanged, with OLS giving lower RMSE and MAE.

DIC**Model comparison scores (training data): DIC, WAIC, and LOO-CV**

model	mean_deviance	penalty_pD	DIC
baseline_beta_N02	3711.268659726393	48.72994154082695	3759.99860126722
alternative_beta_N01	3711.075107566599	48.87392648162762	3759.9490340482266

WAIC

model	waic	se
baseline_beta_N02	3759.8874498788155	100.71764668462626
alternative_beta_N01	3759.538745736574	100.67461436884919

LOO

model	elpd_loo	se_elpd	looic	p_loo
baseline_beta_N02	-1880.0534974658033	50.36067852573391	3760.1069949316066	47.09312489136948
alternative_beta_N01	-1879.8587259934404	50.33884376330298	3759.717451986881	46.971758651291154

LOO COMPARE

model	elpd_diff	se_diff	elpd_loo	se_elpd_loo	p_loo	se_p_loo	looic	se_looic
alternative_beta_prior	0	0	-1879.8587259934404	50.33884376330298	46.971758651291154	2.0390603384629866	3759.717451986881	100.67768752660596
baseline	-0.19477147236443315	0.1480659506870344	-1880.0534974658033	50.36067852573391	47.09312489136948	2.062856384637105	3760.1069949316066	100.72135705146782

Lower DIC, WAIC, and LOOIC indicate better in-sample fit; higher elpd_loo is better. Scores compare baseline vs alternative beta priors on the same training likelihood.

6. Discussion and Conclusion

This project used a Bayesian hierarchical regression model to predict used car prices from vehicle characteristics and brand level variation. The model included fixed effects for age, mileage, fuel type, transmission type, accident history, and clean title status, while allowing baseline prices to vary across brands through random intercepts. This partial pooling structure can stabilize estimates for smaller brand groups.

For this dataset and train-test split, the Bayesian hierarchical model did not outperform the OLS regression baseline in point prediction. OLS achieved lower RMSE and MAE, likely because many major brands had enough observations for ordinary brand indicators to estimate brand differences effectively. In that situation, partial pooling may provide less predictive gain.

The Bayesian model still provided useful information beyond point predictions. Its 95% posterior predictive intervals covered about 94.5% of test observations, close to the nominal level. Therefore, even though the Bayesian model was not better on RMSE or MAE, it remained competitive and gave calibrated uncertainty estimates.

The main limitations are that listed prices may differ from transaction prices, important predictors such as condition, trim, location, options, seller type, and local market conditions were unavailable, and high-priced vehicles were harder to predict. The results are also based on one train-test split, so future work could use cross-validation and richer vehicle features.

Overall, both models produced reasonable predictions. OLS was slightly stronger for point accuracy, while the Bayesian hierarchical model offered a useful framework for grouped data and uncertainty quantification.

Works Cited

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Appendix A: Group Contributions

Xing Sun contributed to data preprocessing, R implementation, JAGS model fitting, MCMC diagnostics, and prediction evaluation.

Ning Xu contributed to exploratory data analysis, data cleaning summaries, variable transformation documentation, and figure/table preparation.

Martin Wang contributed to Bayesian model formulation, posterior predictive interval analysis, results visualization, and drafting the discussion of model limitations.

Jack Liu contributed to data research, project proposal design, prior assessment, model score comparison, and Monte Carlo error analysis.

Appendix B: Use of Generative AI

Generative AI tools were used to assist with brainstorming, grammar checking, and debugging. All analysis, code, results, interpretations, and references were reviewed and verified by the authors, who take responsibility for the final report.

Appendix C: R and JAGS Code

https://github.com/officialjackliu/stat_431_final

Complete R/JAGS code is available in the GitHub repository above. The R Markdown file is located in the report folder.